Lab III Draft

Measurement Protocol

Phase 0 - Material characterization

1. Print six tensile coupons from the same spool you'll use for the experiment.

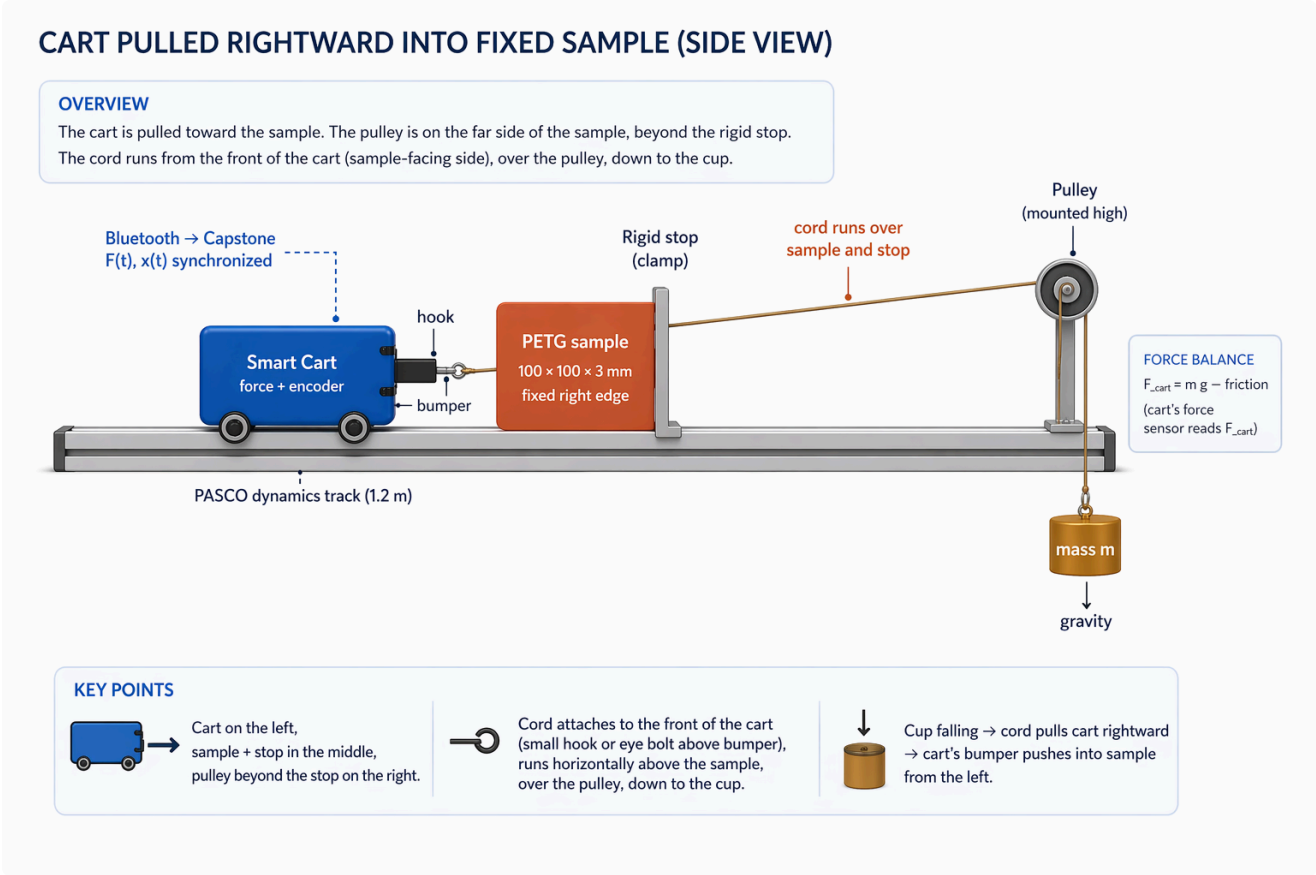
- Three at 0° raster (load direction parallel to last-layer lines)
- Three at 45° (the orientation you'll use for all specimens)

2. Pull the coupons on the PASCO cart rig with the quasi-static pulley loading, or on a dedicated tensile test machine if your lab has one.
3. Extract:
   - E (initial slope)
   - σ_y
   - Record the raster-angle dependence
4. Weigh a known-volume printed cube (20 × 20 × 20 mm, 100% infill) to get effective ρ — this includes any print porosity and will be slightly below the 1270 kg/m³ nominal value.
5. Log these four numbers (E, σ_y, ρ, raster-angle factor).
6. Recompute the target mass:
   $m* = \rho_{measured} \cdot V_f \cdot V_{region}$

Phase 1 - Specimen Preparation (per specimen)

1. CAD the geometry. Verify CAD body volume equals $V_f \cdot V_{region} = 9000mm^3$ in sofrware.
2. Slice at 100% infill, layer height 0.2 mm, 3–4 perimeters, 45° raster, 240/80 °C temperatures, 40 mm/s print speed. Keep these identical across all specimens.
3. Print. Let the specimen cool on the bed for 10 minutes after completion before removing.
4. Inspect for layer adhesion defects, warping, stringing across the gaps, and visible voids. Reject and reprint if any are present.
5. Weigh on the precision balance. Record mass. Reject if outside $m^* \pm 2\%$
6. Label the specimen (topology code, print date, mass).

Phase 2 (once per test session)



1. Power on the PASCO Smart Cart and Capstone software.
2. Level the track with a bubble level on both axes.
3. Attach the cord between cart and pulley, no mass in the cup. Tare the force sensor in Capstone with the cord attached.
4. **Calibration check.** Hang three known masses from the cord: 100 g, 200 g, 500 g. Verify Capstone reads mgmg mg = 0.98, 1.96, 4.90 N respectively, within ±2%. If the error is systematic (e.g. all readings 3% low), record the correction factor and apply in post-processing.
5. **Friction baseline.** Remove the sample stop and run the cart through the full 0–20 mm range of the working zone with no sample present. Record $F(x)$. The residual is your friction + cord drag signature — save this file; you'll subtract it from every subsequent measurement. It should be roughly flat at 0.1–0.3 N. If it's >0.5 N or non-monotonic, re-seat the pulley bearings.
6. Reinstall the rigid stop. Verify the cart's bumper makes contact with the stop plane (no sample) at the position you'll call $x = 0$. Zero the position channel in Capstone at this point.

Phase 3 - H1 Measurement Protocol (Per specimen, 3 repeats)

1. Mount the specimen against the rigid stop. The fixed edge is clamped against the stop; the free edge faces the cart.
2. Connect the data log in Capstone: sample rate 100 Hz, channels force and position, trigger on force > 0.5 N to skip the pre-contact dead zone.

3. **Ramp load.** Gradually pour water into the cup at a rate that takes ~60 seconds to reach the target displacement (20 mm, or the first sign of nonlinearity — whichever comes first). Strain rate: 20 mm / 60 s / 100 mm = 0.003 s^{-1}, solidly quasi-static for PETG.
4. **Ramp unload.** Gradually siphon water out over ~60 seconds. Record the unload curve.
5. **Recovery pause.** Leave the specimen at rest for 2 minutes. PETG is viscoelastic and recovers on this timescale; without the pause, cycle 2 starts from a "warm" state.
6. Repeat load-unload two more times (3 total cycles).
7. Save the Capstone file as `H1_<topology>_<specimen_id>.csv` containing t, F, x for all three cycles.

Phase 4 - H2 Measurement Protocol (Per specimen)

1. Mount a specimen from each of the TO variants plus the best two H1 baselines (honeycomb, bistable) as controls.
2. **Preconditioning.** Run 3 slow cycles to $0.5x_y$ (pour and siphon over ~30 s each direction).
3. **Cyclic loading, 100 cycles.** Each cycle: load to 0.5 x_y, hold 2 s, unload to 0 , hold 2 s. Automate or use metronome-paced manual pour/siphon at 30 s period.

Hypothesis

H1: Structure Determines Specific Energy Within a Fixed region and mass budget

Among four candidate structure class in the identical 100 x 100 x 3 mm region at matched mass:
(a) cantilever beam
(b) Archimedean spiral
(c) honeycomb lattice
(d) passive bistable arch array.
Structures that operate through distributed bending or post-buckling store one/two orders of magnitude more energy per unit mass than slender beam topologies.

Symbol	Quantity	Type	Measured by
L, t	Region dimensions (100 x 100, 3 mm)	Controlled	Print Geometry
V_f	Material Volume Fraction = 0.30 $V_f = \frac{V_{\text{material}}}{V_{\text{region}}}$	Controlled	
E, σ_y, ρ	PETG Young's modulus, yield stress, PETG density	Material Property	Coupon test per print batch

Symbol	Quantity	Type	Measured by
Geometry class	{beam, spiral, honeycomb, bistable}	Independent (categorical)	4 STL Models
$x(t)$	Applied Displacement	Controlled (quasi-static)	PASCO Smart Cart encoder, ~0.2 mm resolution
$F(x)$	Reaction Force	Dependent (primary)	PASCO bumper force sensor, $\pm 50N, \pm 0.1N$
$U = \int_0^{x^*} F dx$	Stored energy up to usable displacement	Dependent (derived)	Capstone Integral
m	Specimen mass	Controlled/measured	Precision balance
U/m	Specific energy	**Primary outcome**	U/m
x_y	Yield Displacement	Dependent	First 5% deviation from linear fit
x^*	Test endpoint	Define	0.8 x_y (linear desings) or first snap (nonlinear)

Volume Fraction V_f

Ratio of material volume to the total enclosed design region. Defines the target mass of material (for example, PETG) per specimen. The other space is empty space.
It is controlled because

1. Decouples topology from "how much material you threw into it". A heavy design always store more energy than a light one in absolute terms. Fixing V_f means difference in stored energy between specimens is solely due to **how** the material is arranged, not **how much** there is.
2. It is a canonical TO constraint. SIMP algorithm is literally asked "given that you may use a fraction V_f of the region, distribute it to minimize compliance.
3. $V_f = 0.30$ is the standard benchmark value.

Verification: weight every specimen. Target mass $m^* = \rho \cdot V_f \cdot V_{region}$

H2 - Within TO, objective formulation determines performance more than the fact of optimizing

TO = algorithmns that input

- Design region
- Material Budget
- Objective function (for example, compliance minimization, displacement maximization) and outputs the layout that best satisfies that objective.

The default story in introductory TO papers and undergraduate projects goes like this:

> "We ran topology optimization on our design problem. The optimized structure performs X% better than a hand-designed baseline. Therefore, TO is useful."

This is a weak claim, because it compares two variables at once — (a) whether you optimized, and (b) what you optimized for — and attributes all the improvement to (a). It's like saying "I tuned my car and it's faster, therefore tuning helps" without noting that you also switched from racing fuel to premium.

We want to test if compliance-minimizing TO is dominated by a well-chosen baseline (like the honeycomb or bistable array from H1)

Symbol	Quantity	Type
λ	TO objective weight	Indp
Geometry class	Comparison category {SIMP-TO, buckling-TO, honeycomb, bistable}	Indp

Dependent variables for H2

Symbol	Variable	How measured	Why it matters
$F(x)$	Force–displacement curve	PASCO cart bumper force sensor + wheel encoder, 100 Hz, logged in Capstone	The raw data; every other quantity derives from this
U	Stored energy	$U = \int_0^{x^*} F\,dx$ via trapezoidal integration	Total elastic energy the structure holds before yield or snap
U/m	Specific stored energy	U divided by measured specimen mass (J/kg)	**Primary comparison metric** — efficiency of material use
x_y	Yield displacement	0.2% offset method from the initial linear fit	Defines the usable loading range; sets the H2 cycle amplitude at $0.5\,x_y$
F_{peak}	Peak force	Maximum of $F(x)$	Distinguishes linear designs (monotonic F) from snap-through designs (peak then descent)
η_N	Fractional hysteresis per cycle N	Loop area $\oint F\,dx$ divided by loading-branch energy	**Primary H2 metric** — measures irreversibility (energy lost per cycle)
η_{plateau}	Plateau hysteresis	$\langle \eta_N \rangle$ averaged over cycles 10–100	Post-break-in steady-state hysteresis; the meaningful cyclic-stability number

Symbol	Variable	How measured	Why it matters
k_N	Loading-branch stiffness per cycle	Linear fit slope over $[0.2,\ 0.6]\,x_{\max}$	Tracks cycle-to-cycle softening from damage accumulation
$\Delta k/k_0$	Stiffness drift over 100 cycles	$(k_{100} - k_1)/k_1$	Secondary metric; quantifies structural degradation under repeated loading

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